

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- No recommended Cummins® service tools required

Additional Service Items

- Non-metallic brush
- Feeler gauge.

Preparatory Steps

WARNING

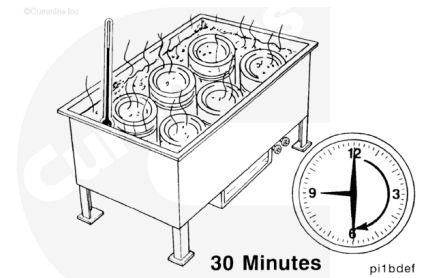
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the lubricating oil system. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/1016/1016-007-037.html>)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html>)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/1016/1016-002-004.html>)
- Remove the oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html>)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/1016/1016-001-089.html>)
- Remove the piston cooling nozzle. Refer to Procedure 001-046 in Section 1. (</qs3/pubsys2/xml/en/procedures/1016/1016-001-046.html>)
- Remove and disassemble the piston and connecting rod assembly. Refer to Procedure 001-054 in Section 1. (</qs3/pubsys2/xml/en/procedures/1016/1016-001-054.html>)

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ CAUTION ⚠

Do not use the bead blast method to clean the piston. The piston will be damaged by blast material embedded in the aluminum.

⚠ CAUTION ⚠

Do not clean the pistons and connecting rods in an acid bath.

⚠ CAUTION ⚠

Do not use a metal brush. A metal brush will damage the piston ring grooves.

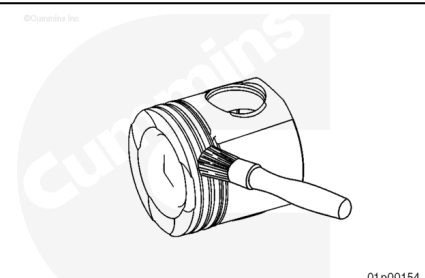
It is **only** necessary to remove the carbon buildup on the piston crown. It is **not** necessary to make the crown appear to be new.

Soak steel pistons for at least 30 minutes in an approved cleaning solvent for steel. Use a non-metallic brush to remove carbon deposits.

For best results, soak the pistons for several hours or overnight.

⚠ CAUTION ⚠

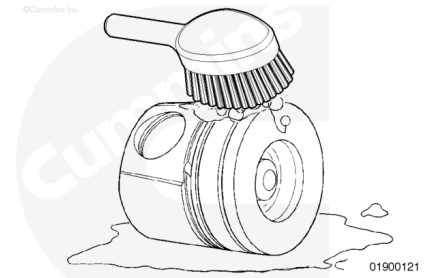
Do not use a ring groove cleaner to clean the ring grooves.
Do not scratch the ring sealing surface in the piston groove.



Clean the first and second compression ring grooves. Use solvent and a brush with plastic bristles.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

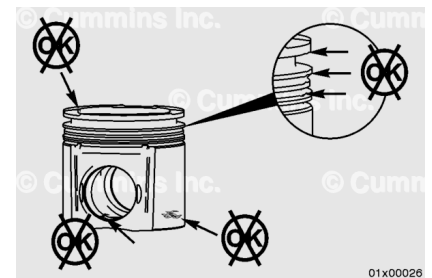


Wash the pistons again. Use an approved cleaning solvent for steel.

Dry with compressed air.

⚠ CAUTION ⚠

Do not use pistons with cracks. If the crown is cracked, the piston must be replaced. Failure to replace the damaged piston can cause serious engine damage.



Failure Analysis Inspection

Inspect the piston for damage and excessive wear. Check the top, ring grooves, skirt, and pin bore for cracks or damage.

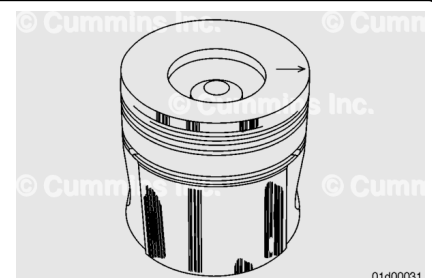
If significant damage is found, the piston **must** be replaced.

Note : If severe piston damage has occurred, check the turbocharger and other exhaust components for damage from debris.

Abrasive and/or Debris Wear

Scratching on the piston skirt with material embedded in the piston can be caused by:

- Ingested abrasive material
- Inadequate cleaning during a previous repair
- Particles embedded in the cylinder bore
- Improper maintenance of the lubricating oil system
- Debris in the lubricating oil system from another malfunction
- Scuffing and scoring.



If the scratches can be felt with a fingernail, do **not** reuse the piston.

Scuffing and scoring can be caused by:

- Engine overheating

- Oil dilution
- Improper maintenance of the lubricating oil system
- Piston cooling nozzle malfunction
- Oil ring groove plugged by deposits
- Injector over fueling or running engine on an alternative fuel source, i.e. oil in the intake system.

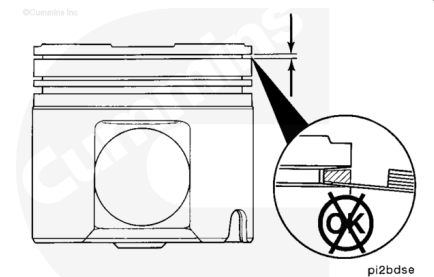


The piston ring groove can be inspected with a new piston ring and a feeler gauge.

Hold a new piston ring in the groove of the piston.

Install a 0.15 mm [0.006 in] feeler gauge.

If the feeler gauge enters the groove without resistance, there is too much wear. Replace the piston.



Other Types of Piston Malfunctions

Hydraulic Lock

Hydraulic locks (oil, fuel, or water) can cause the piston to split in half, or the lands can break and the connecting rods can be bent.

Dustouts

Particular attention to the wear pattern of the intermediate (second) ring is required for diagnosis of a dustout piston scuff.

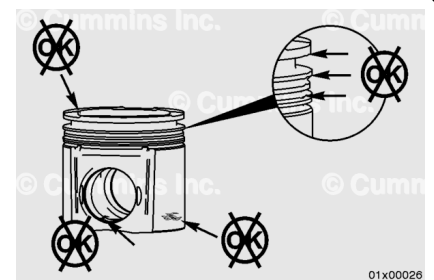
The ring will have a polished appearance on the full-face ear of the intermediate ring, where the blackened areas of the ring become more polished.

The polished area starts at the bottom of the ring and continues upward on the tapered ring.

Occasionally, the edges of the rings are razor sharp, and the block cylinder bore has indications of ridges approximately 1-½ inches below the top deck.

Top of Piston Meltdown

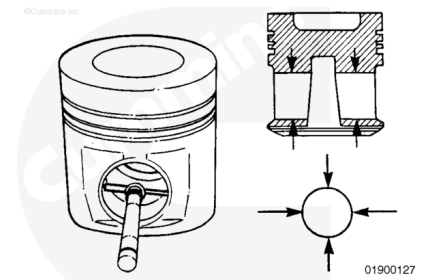
The top of the piston appears to be melted down. The malfunction resembles an overfueling situation such as a stuck injector. A plugged piston cooling nozzle can resemble this malfunction as well.



Measure

Measure the piston pin bore when the piston temperature is 20°C [68°F].

Piston Pin Bore Diameter		
mm		in
52.09	MIN	2.051
52.11	MAX	2.052

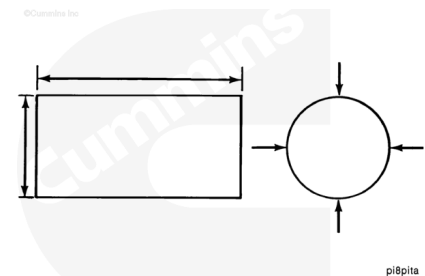


Note : Add 0.013 mm [0.0005 in] to the bore diameter per 5°C [9°F] temperature rise up to 32°C [90°F].

Inspect the piston pin for nicks, gouges, and excessive wear.

Measure the piston pin outside diameter.

Piston Pin Diameter		
mm		in
51.0065	MIN	2.008
28.0035	MAX	2.047



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Assemble the piston and connecting rod assembly. Refer to Procedure 001-054 in Section 1. (</qs3/pubsys2/xml/en/procedures/1016/1016-001-054.html>)
- Install the piston cooling nozzle. Refer to Procedure 001-046 in Section 1. (</qs3/pubsys2/xml/en/procedures/1016/1016-001-046.html>)
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/1016/1016-001-089.html>)
- Install the oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html>)

- Install the cylinder head. Refer to Procedure 002-004 in Section 2.
(/qs3/pubsys2/xml/en/procedures/1016/1016-002-004.html)
- If malfunction results in coolant, oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected.
- Fill the lubricating oil system. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/1016/1016-007-037.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.

Note : All engines **must** be run-in after a rebuild or a repair involving the replacement of one or more piston ring sets, cylinder liners, or pistons. See Section 14 (Engine Testing) for the appropriate run-in procedure.

- Use the following procedure for a general run-in test overview. Refer to Procedure 014-999 in Section F. (/qs3/pubsys2/xml/en/procedures/1016/1016-014-999.html)